

LostLoop

Project specification

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Project overview

LostLoop is an AI lost-and-found prototype that connects a finder's photo with an owner's description and guides verification before a proposed handover. Hossam Elshahaby built the project for the Agentic AI Hack in Stockholm on 3 October 2026. This specification was prepared on 4 October 2026.

Problem and users

Lost items create repeated searches and manual comparison work. Owners often lack a single place to describe a loss. Reception teams need evidence before returning an item. Initial users are event participants, finders and reception staff. Campuses and workplaces are potential pilot settings.

Current prototype evidence

The public homepage, found-item form and claim interface were inspected. The form exposes photo upload, demo photos, title, category, colour, description, location and a hidden identifying detail. The claim screen attributes the agent to Gemini and displays a live activity log. In one demo run, the agent searched two candidates and asked about the backpack's contents. The homepage labels the backpack as demo data. Photo analysis, completed verification, notifications, persistence and handover execution were not independently validated.

Links

Live app: <https://itemlostandfound.lovable.app>

Demo video: https://youtu.be/P0A6O4n_xd8

Functional requirements

- FR1: A finder can submit an image and edit the draft listing before publishing.
- FR2: Public listings show useful attributes while withholding verification details.
- FR3: An owner can describe an item in ordinary language.
- FR4: The agent searches available records and asks for clarification when candidates are ambiguous.
- FR5: The agent asks an ownership question without disclosing its expected answer.
- FR6: An authorized custodian confirms any proposed handover and final return.
- FR7: Each case records its actions and status changes.

Agent workflow

Receive description; search records; assess candidates; ask a clarification or verification question; evaluate the answer; propose a handover only after sufficient evidence; record custodian confirmation. An unresolved or inconsistent claim remains open for human review. A similarity score represents matching evidence, not proof of ownership.

Proposed architecture

Recommended design: a web client connects to an authenticated backend that stores listings and claim sessions. A Gemini agent calls constrained tools for search, verification and handover proposals. Private verification fields remain server-side. Image storage and the database enforce access permissions. This is a proposed implementation design, not a confirmed description of the deployed code.

Data and permissions

Suggested records: Item (id, photo reference, public description, category, colour, venue, status); PrivateDetail (item id, restricted answer or evidence); Claim (id, item id, session, evidence, decision); Handover (claim id, custodian, proposed slot, confirmation); AgentAction (claim id, action, timestamp, outcome). Keep private answers out of public endpoints and logs. Model instructions alone must not enforce access control.

Verification and limitations

A single hidden detail can be guessed or leaked. The public demo intentionally reveals the expected contents so presenters can complete the example. Production cases must never publish those hints. Use additional evidence and custodian review for valuable items. Rate-limit guesses and isolate claims so one user cannot inspect another user's private information. Treat listing text as untrusted input.

Acceptance tests

1. Submit a disposable test item and confirm public fields while private details stay absent from public views.
2. Search a known item and confirm the tool log reflects the actual records searched.
3. Give an incorrect ownership answer and confirm no handover is authorized.
4. Use ambiguous descriptions and confirm the agent asks for clarification.
5. Confirm a correct claim with a custodian and verify the final returned status persists after refresh.
6. Check unauthorized access to private fields and cross-user claims is denied.
7. Simulate a model or database failure and confirm the UI gives an honest retry state.

Impact evaluation and roadmap

Expected benefits are faster recovery and fewer repetitive staff interactions. No recovery-rate, cost-saving or accuracy results are established. A pilot should measure median recovery time, custodian-confirmed returns, incorrect approvals and staff minutes per case. Next steps: complete end-to-end tests, strengthen private-detail handling, add custodian review and pilot with one venue before expanding.

Demo script

Open the app and choose "I lost something". Enter "I lost a black backpack with a red keychain". Observe record search and the verification question in the activity panel. For the labelled demo only, the homepage supplies the expected contents: a laptop and a blue notebook. Explain that production verification answers must stay private. Use the supplied video if network access interrupts the live presentation.